90MW FLOATING SOLAR PV POWER PROJECT IN OMKARESHWAR

Document Prepared by Kosher Climate India Private Limited

Project title	90MW FLOATING SOLAR PV POWER PROJECT IN OMKARESHWAR
Project ID	5034
Crediting period	12-April-2024 to 11-April-2031
Original date of issue	27-March-2024
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Version	02
VCS Standard Version	4.6
Prepared by	Kosher Climate India Private Limited

CONTENTS

1 PROJECT DETAILS

1.1 Summary Description of the Project

The purpose of this project is to generate electricity by harnessing the solar energy by using of solar photovoltaic technology and there by feed the generated electricity to the Indian national grid. The solar photovoltaic based panels convert the available solar radiation into the DC power and there by installed the Inverters to convert the DC power to the AC Power.

SJVN Green Energy Limited (A Wholly Owned Subsidiary of SJVN limited) to develop a 90 MW floating solar PV plant on the surface of Omkareshwar Dam at Punasa Tehsil, located in the village of Kheda, Bilaya-Ekhand, Khandwa district, Madhya Pradesh, India.

The project was awarded to SJVN Limited by Rewa Ultra Mega Solar Limited (RUMSL) through competitive bidding process. Electricity generated from the project is evacuated to national grid of India. MP Power Management Company Limited (MPPMCL), a wholly owned company of MP Government, is procuring all the units generated from the project. A long-term power purchase agreement (PPA) has been executed between SJVN Green Energy Limited and MPPMCL & RUMSL.

The scenario existing prior to the implementation of the project is electricity delivered to the grid by the project that would have otherwise been generated by the operation of grid connected power plants and by the addition of new generation sources, as reflected in the combined margin (CM) calculations.

Project activity is expected to generate annual average electricity of 177,689 MWh and reduces an annual average emission reductions equivalent to 166,175 tCO_2e and 1,163,222 tCO_2e over the crediting period of 7 years.

1.2 Audit History

Audit type	Period	Program	Validation/verification body name	Number of years
Validation/ verification	validation	VCS	Enviance Service Private Limited	07 years

1.3 Sectoral Scope and Project Type

Sectoral scope[1]	Scope: 01 - Energy (renewable / non-renewable)
Project activity type	Type I – Renewable Energy Projects

[1] *Projects, activities, or methodologies may be developed under any of the 16 VCS sectoral scopes:* https://verra.org/programs/verified-carbon-standard/vcs-program-details/#sectoral-scopes

1.4 Project Eligibility

1.4.1 General eligibility

As per section 2.1.3 of VCS Standard version 4.6, table no. 1 excluded project activities (The exclusion does not apply to concentrated solar thermal to electricity, floating solar PV, or energy storage systems e.g., batteries).

The project activity applies the large-scale CDM approved methodology ACM0002: Grid-connected electricity generation from renewable sources, Version 21.0 which is eligible under VCS GHG program. Further, this project is not a fragmented part of a larger project or activity that would otherwise exceed such limits.

Thus, the proposed project is eligible under the scope of the VCS program.

1.4.2 AFOLU project eligibility

This is a non-AFOLU project. This section is not applicable.

1.4.3 Transfer project eligibility

This project is not registered under any other GHG programs and it is not a CPA project. Hence, this section is not applicable.

1.5 Project Design

☒ Single location or installation

☐ Multiple locations or project activity instances (but not a grouped project)

☐ Grouped project

1.5.1 Grouped project design

This is not a Grouped project. Hence, Not Applicable.

1.6 Project Proponent

Organization name	SJVN Green Energy Limited
Contact person	Pushkar Verma
Title	DGM/EIC
Address	SJVN Corporate Head Quarters, Shakti Sadan, Shimla - 171006

Telephone	+91 9418450295
Email	pushkar.verma@sjvn.nic.in

1.7 Other Entities Involved in the Project

Organization name	Kosher Climate India Private Limited
Role in the project	Project Representative
Contact person	Vamsi Krishna Manchikalapudi
Title	Founder & Managing Director
Address	Zee Plaza, No.1678, Ground Floor, 27th Main Road, Sector 2, HSR Layout, Bengaluru, Karnataka-560102, India
Telephone	+91 9945343475
Email	narendra@kosherclimate.com, vamsi@kosherclimate.com

1.8 Ownership

SJVN Green Energy Limited is the legal owner of the project activity. Respective project commissioning certificate and the Power Purchase Agreement of the project would be the supporting document to demonstrate the project ownership. This demonstrates the right of use according to clause 3.7.1 (3) of VCS Standard (v4.6) – "a project ownership arising by virtue of a statutory, property or contractual right in the plant, equipment or process that generates GHG emission reductions and/or removals". Also, other legal compliances may be considered.

1.9 Project Start Date

Project start date	12-April-2024 (Expected)
Justification	As per the section 3.8 of the VCS standard v4.6, the project start date of a non-AFOLU project is the date on which the project began generating GHG emission reductions or carbon dioxide removals. Since the project is not yet started the commercial operations, the expected start of commercial operations has been considered as the project start date.

1.10 Project Crediting Period

Crediting period	☒ Seven years, twice renewable

	☐ Ten years, fixed ☐ Other (state the selected crediting period and justify how it conforms with the VCS Program requirements)
Start and end date of first or fixed crediting period	12-April-2024 to 11-April-2031

1.11 Project Scale and Estimated GHG Emission Reductions or Removals

☒ < 300,000 tCO2e/year (project)

☐ ≥ 300,000 tCO2e/year (large project)

The table below shows the Estimated GHG emission reductions or removals (tCO_2e) for the fixed crediting period:

Calendar year of crediting period	Estimated GHG emission reductions or removals (tCO_2e)
12-Apr-2024 to 31-Dec-2024	122,175
01-Jan-2025 to 31-Dec-2025	168,073
01-Jan-2026 to 31-Dec-2026	167,233
01-Jan-2027 to 31-Dec-2027	166,397
01-Jan-2028 to 31-Dec-2028	165,565
01-Jan-2029 to 31-Dec-2029	164,737
01-Jan-2030 to 31-Dec-2030	163,913
01-Jan-2031 to 11-Apr-2031	45,130
Total estimated ERRs during the first or fixed crediting period	**1,163,222**
Total number of years	**7**
Average annual ERRs	**166,175**

1.12 Description of the Project Activity

The purpose of this project is to generate electricity by harnessing the solar energy by using of solar photovoltaic technology and there by feed the generated electricity to the Indian national grid. The Solar photovoltaic based panels convert the available solar radiation into the DC power and there by installed the Inverters to convert the DC power to the AC Power.

SJVN Green Energy Limited (A Wholly Owned Subsidiary of SJVN limited) to develop a 90 MW floating solar PV plant on the surface of Omkareshwar Dam at Punasa Tehsil, located in the village of Kheda, Bilaya-Ekhand, in the Indian state of Madhya Pradesh.

The scenario existing prior to the implementation of the project is electricity delivered to the grid by the project that would have otherwise been generated by the operation of grid connected power plants and by the addition of new generation sources, as reflected in the combined margin (CM) calculations.

The generated electricity from the project is supplied to the Indian National grid. Project Owner has signed a Power Purchase Agreement with MPPMCL to supply the generated solar power at a fixed price for a period of 25 years.

Project Name	Capacity (MW)	Purpose	Commissioning Date (COD)	Village
90MW FLOATING SOLAR PV POWER PROJECT IN OMKARESHWAR	90	Sale to MPPMCL	12/04/2024 (Expected)	Kheda, Bilaya-Ekhand

Project activity is expected to generate annual average electricity of 177,935 MWh and reduces an annual average emission reductions equivalent to 166,405 tCO_2e and 1,164,835 tCO_2e over the crediting period of 7 years.

Technologies/measures

Project used Crystalline Photovoltaic technology which converts the solar radiation into the electricity. The solar PV plant has the PV modules, Central Inverters, Transformers and other relay and protection systems.

Technical specifications of the major components of the project activity are given below. Solar PV modules were Seasonal tilt mounted and overall life of the projects are 25 years.

Parameters	**90 MW Floating Solar**
Project Capacity	Capacity- 90 MW (AC)
Technology	Bifacial crystalline Module
Module Make	Longi HK trading Limited
Module Model Number	LR5-72HBD-540M, LR5-72HBD-545M, LR5-72HBD-550M
Total Number of Modules	540Wp-95,040, 545Wp-117,660, 550Wp-6,750
Module efficiency	21.5%
Inverter Make	Sineng electric (INDIA)Pvt Ltd
Inverter	**Capacity** - 3.3 MVA
	Number of Inverters - 28 Nos.
Inverter Duty Transformer Make	Telawne power equipments private limited
Capacity	6.6MVA
Mounting Structure	High Density Polyethylene (HDPE) Floats

Lifetime of the project	25 Years

Solar PV modules converts the available solar radiation into the DC power. Installed Central Inverters will convert the generated DC power into the 660 V AC Power. Total AC power from all the inverter blocks will be pooled into the common switchyard. Total Power will be stepped up to 33 kV in the switchyard by using step-up transformers and transmitted to the nearest substation.

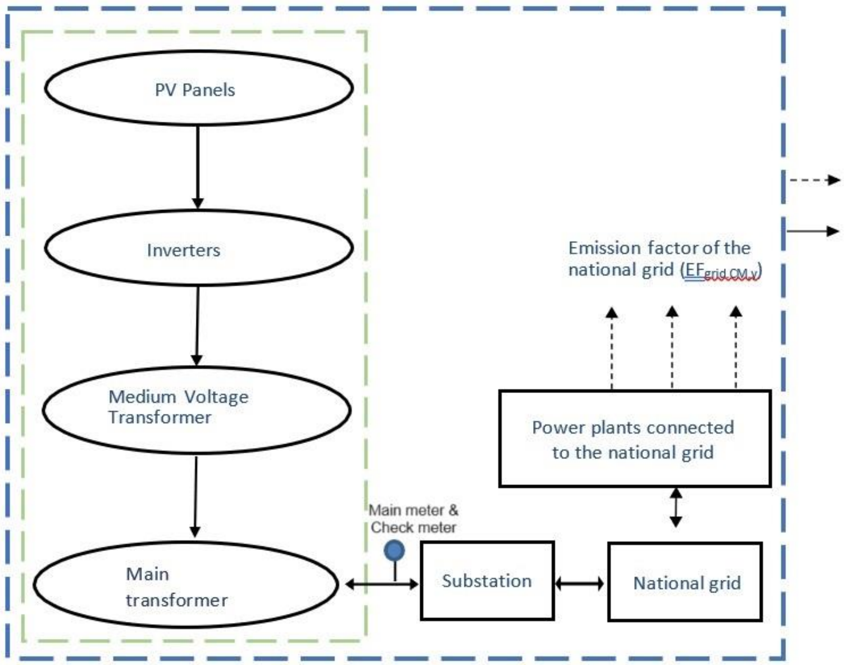


Figure 2: Solar power plant flow diagram

An intelligent automatic monitoring and alarm system (SCADA-Supervisory Control and Data Acquisition) has been already installed in the project control room which will monitor and record the real time data from the plant and alerts the staff in case of any malfunctioning in the equipment operation. However, Separate Energy meters will be installed by MPTRANSCO to record the import and export of electricity from the plant.

This is a Greenfield project activity generating the electricity from the solar energy and supplying it to the national grid. In the baseline scenario the equivalent of electricity delivered to the grid by the project activity would have otherwise been generated by the operation of grid-connected power plants and by the addition of new generation sources, as reflected in the combined margin (CM) calculations described in the Tool 07 “Tool to calculate the emission factor for an electricity system”, version 7.0. There is no technology transfer occurred in the project activity. There was no activity at the site of the project owner prior to the implementation of this project. Hence pre-project scenario and baseline scenario is the same

1.13 Project Location

The Project is located in the state of Madhya Pradesh, India.

Address and geographic coordinates of the physical site of the project			
Project	**Physical address**	**Latitude**	**Longitude**
90mw Floating Solar PV Power Project in Omkareshwar	Omkareshwar Dam, Khandwa, Madhya Pradesh, India	22°14'25''N (22.2402°N)	76°09'45''E (76.1625°E)

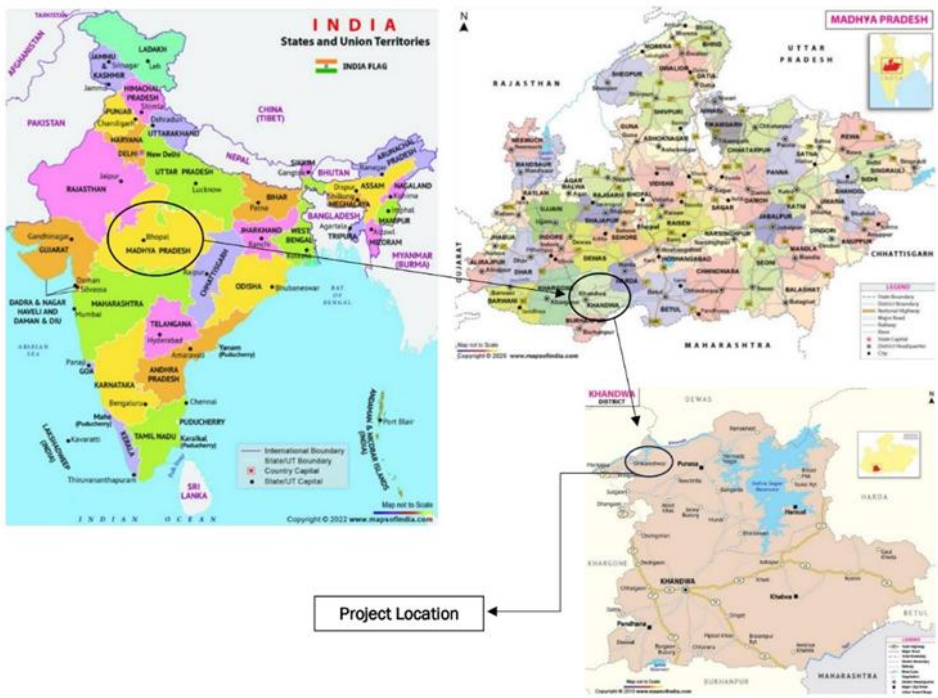


Figure: Project Location

1.14 Conditions Prior to Project Initiation

The scenario existing prior to the implementation of the project activity, is electricity delivered to the grid by the project activity that would have otherwise been generated by the operation of grid connected power plants and by the addition of new generation sources, as reflected in the combined margin (CM) calculations described in the "Tool 07: Tool to calculate the emission factor for an electricity system", version 7.0. This is a Greenfield project activity. There was no activity at the site of the project owner prior to the implementation of this project activity. Hence pre-project scenario and baseline scenario is the same.

Refer Section 3.4 (Baseline Scenario) for detailed understanding of the same.

1.15 Compliance with Laws, Statutes and Other Regulatory Frameworks

The project conforms to all the applicable laws and regulations in India:

- Power generation using renewable energy is not a legal requirement or a mandatory option.
- The Indian Electricity Act, 2003 (May 2007 Amendment) does not influence the choice of fuel used for power generation.
- There is no legal requirement on the choice of a particular technology for power generation

There is no mandatory requirement to implement the project. The project comes under white category as per local regulation, thus there shall be no necessity of obtaining the "Consent to Operate" for White category of industries. Since project falls under white category and the non-polluting nature of project fulfils the compliance to the local laws and regulations.

As per the paragraph no 3.1.4 of the VCS standard v4.6 the project conforms to all the applicable laws and regulations and is not implemented by the force of law. This is a voluntary activity undertaken by the project owner in compliance with all the legal requirement in the host country.

1.16 Double Counting and Participation under Other GHG Programs

1.16.1 No Double Issuance

Is the project receiving or seeking credit for reductions and removals from a project activity under another GHG program?

☐ Yes ☒ No

1.16.2 Registration in Other GHG Programs

Is the project registered or seeking registration under any other GHG programs?

☐ Yes ☒ No

1.16.3 Projects Rejected by Other GHG Programs

Has the project been rejected by any other GHG programs?

☐ Yes ☒ No

1.17 Double Claiming, Other Forms of Credit, and Scope 3 Emissions

1.17.1 No Double Claiming with Emissions Trading Programs or Binding Emission Limits

Are project reductions and removals or project activities also included in an emissions trading program or binding emission limit?

☐ Yes ☒ No

1.17.2 No Double Claiming with Other Forms of Environmental Credit

Has the project activity sought, received, or is planning to receive credit from another GHG-related environmental credit system?

☐ Yes ☒ No

1.17.3 Supply Chain (Scope 3) Emissions

Do the project activities specified in Section 1.12 affect the emissions footprint of any product(s) (goods or services) that are part of a supply chain?

☐ Yes ☒ No

Is the project proponent(s) or authorized representative a buyer or seller of the product(s) (goods or services) that are part of a supply chain?

☐ Yes ☒ No

Has the project proponent(s) or authorized representative posted a public statement on their website saying, "Carbon credits may be issued through Verified Carbon Standard project [project ID] for the greenhouse gas emission reductions or removals associated with [project proponent or authorized representative organization name(s)] [name of product(s) whose emissions footprint is changed by the project activities]."

☐ Yes ☒ No

1.18 Sustainable Development Contributions

The project is expected to contribute 03 SDGs which are SDG 7, 8 and 13.

1. Affordable and clean energy (SDG-07)

Goal: Ensure access to affordable, reliable, sustainable and modern energy for all

Target: 7.2: By 2030 increase substantially the share of renewable energy in the global energy mix

Indicator: 7.2.1: Renewable energy share in the total final energy consumption

The project activity generating 90MW of electricity through solar energy which would have been generated from the grid connected power plants.

Monitoring Parameter: Net Electricity Generation (GWh/year)

2. Decent work and economic growth (SDG-08)

Goal: Promote sustained, inclusive and sustainable economic growth, full and productive employment and decent work for all

Target: 8.5 By 2030, achieve full and productive employment and decent work for all women and men, including for young people and persons with disabilities, and equal pay for work of equal value

Indicator: 8.5.1 Average hourly earnings of employees, by sex, age, occupation and persons with disabilities

The project activity provides employment opportunities and the average salaries has been paid to employees irrespective of sex, age group, etc which leads to increased productivity, driving economic development.

Monitoring Parameter: Number of employment (with bifurcation on number by sex, age group and where applicable, persons with disabilities), Average earnings, Policy for Non-discrimination and equal pay for the work of equal value.

3. Climate action (SDG-13)

Goal: Take urgent action to combat climate change and its impacts

Target: 13.2 Integrate climate change measures into national policies, strategies and planning

Indicator: 13.2.2 Total greenhouse gas emissions per year

Reduction of annual average of 166,405 tons of CO_2 eq. greenhouse gas (GHG) emissions per year by coal replacement and avoiding open dumping of waste.

Project activity generates renewable energy-based electricity and mitigates the CO_2 emissions which would have been generated from the grid-connected power plants. Project has already commissioned and started reducing the emissions. Hence, complied to the SDG 13.

Monitoring Parameter: GHG emission reductions (tCO_2/year).

1.19 Additional Information Relevant to the Project

1.19.1 Leakage Management

There is no leakage emission involved in the project. Thus, leakage management is not required in this case.

1.19.2 Commercially Sensitive Information

No commercial sensitive information has been excluded from the public version of the project description.

1.19.3 Further Information

Not applicable.

2 SAFEGUARDS AND STAKEHOLDER ENGAGEMENT

2.1 Stakeholder Engagement and Consultation

2.1.1 Stakeholder Identification

Stakeholder Identification	The stakeholders for the project have been prioritized by identifying direct (those who have a direct interest or influence on the project) and indirect stakeholders whose interest is indirect. Stakeholders has been identified based on potential impact from all related facilities associated with the project. Hence, the stakeholders identified are Local people, employees, Local authorities, drivers, etc.
Legal or customary tenure/access rights	The legal and customary rights of the land use and water body use during and post implementation of the project activity lies with The New and Renewable Energy Department, Government of Madhya Pradesh (GoMP-NRE). Project proponent have no rights on the legal ownership of the water body and land. There are no identified disputes or conflicts over rights of the water body and its resources. No concerns were received against the implementation of the project activity and confirms the consent and full support for the project implementation.
Stakeholder diversity and changes over time	Each community has its own cultural practices, customs, and beliefs, contributing to the overall cultural heritage and integrity of the region. The project anticipates no significant changes to the group's cultural integrity in the project region. The project activity by contributing to the environmental, economic, and social benefits, play a crucial role in enabling local communities to remain located in the project area.

Expected changes in well-being	As the project activity leads to the GHG emission reduction through electricity generation from solar plant, the baseline would have involved the grid-connected electricity which is dominated by the fossil-fuel energy sources. Hence, the project activity engages in the emission reductions which enhances the well-being of the environment and local communities.
Location of stakeholders	All community members, those who are directly or indirectly impacted by the project are mostly local people who reside near the project areas.
Location of resources	The location of territories and resources to which stakeholders have customary access is the nearby land and Omkareshwar Dam in Khandwa District of Madhya Pradesh.

2.1.2 Stakeholder Consultation and Ongoing Communication

Date of stakeholder consultation	28-December-2023
Stakeholder engagement process	The LSC invitations were put on public notice to the local people. The consultation attendees were given sufficient time and scope to discuss the project and clarify their doubts related to the project and provide their feedbacks on the project.
Consultation outcome	The PP has conducted a discussion meeting on the project particulars with stakeholders. These sessions ensured unbiased dissemination of all project-related information to stakeholders, irrespective of caste, religion, gender, race, etc., This approach aimed to eliminate any potential barriers hindering the participation of diverse stakeholder groups. The discussion covered a wide range of topics, including project design, location, objectives, costs, benefits, risks, duration, scope, size, impacts, and the carbon project cycle, encompassing validation and verification processes. Based on this information, interested stakeholders provided their inputs through open discussions during the meeting. No negative comments have been received on project activity from any of the local stakeholders consulted. As all comments were very positive about the project, no further action is required. There were no further comments raised by the stakeholders and they were

	totally in support for setting up of these kinds of projects in the region.
Ongoing communication	A dedicated email address as well as the phone number and the address of PP were provided to all stakeholders to foster open and accessible communication. A grievance register will be maintained at the site to register the complaints of the stakeholders. There will be an active presence of the Project Proponent team in the project area. They will work closely with the community and other stakeholders to address any queries or grievances in relation to the project. All communication and consultation shall be performed in a culturally appropriate manner, including language and gender sensitivity.
Stakeholder input	No negative comments have been received on project activity from any of the local stakeholders consulted. As all comments were very positive about the project, no further action is required.

2.1.3 Free Prior and Informed Consent

Obtaining consent	The legal and customary rights of the land use and water body use during and post implementation of the project activity lies with The New and Renewable Energy Department, Government of Madhya Pradesh (GoMP-NRE). Project proponent have no rights on the legal ownership of the water body and land. There are no identified disputes or conflicts over rights of the water body and its resources. No concerns were received against the implementation of the project activity and confirms the consent and full support for the project implementation.
Outcome of FPIC	The legal and customary rights of the land use and water body use during and post implementation of the project activity lies with The New and Renewable Energy Department, Government of Madhya Pradesh (GoMP-NRE). The project proponent has not done encroachment and relocated people, and forced physical or economic displacement.

2.1.4 Grievance Redress Procedure

Development process	The project team devised a structured process in a culturally appropriate manner for receiving input and addressing grievances, by maintaining a grievance register at the site to register the complaints of the stakeholders.
Grievance redress procedure	There is a mechanism in place to address the grievances and resolve any potential conflict between the project proponent and local stakeholders. The project proponent contact information like e-mail and phone numbers are provided to the stakeholders in case they want to contact and share their ideas and convey their grievances.

2.1.5 Public Comments

The Section will be updated post the public comment period.

2.2 Risks to Stakeholders and the Environment

	Risks identified	Mitigation or preventative measure taken
Risks to stakeholder participation	No risks identified	The Project team has explained about the project to the stakeholders including project benefits, costs, risks, rights, laws and opportunities on employment generation. The stakeholders are voluntarily willing to participate in the project because of benefits like safeguarding the communities from environmental challenges and various other SDG benefits of the project.
Working conditions	No risks identified	The project has taken comprehensive measures to ensure providing safe and healthy working conditions for its employees.
Safety of women and girls	No risks identified	The women are provided employment opportunities in the project activity with attention given to addressing gender-specific safety concerns. The organization does not tolerate harassment for any women employees which includes sexual, psychological or any other form of harassment that can be a form of discrimination.

Safety of minority and marginalized groups, including children	No risks identified	The project proponent has protocols aimed at safeguarding women, children and other marginal or vulnerable groups.
Pollutants (air, noise, discharges to water, generation of waste, release of hazardous materials)	No risks identified	The project activities do not involve generation of pollutants.

2.3 Respect for Human Rights and Equity

2.3.1 Labor and Work

Discrimination and sexual harassment	The organization has policies for non-Discrimination and prevention of sexual harassment and would take appropriate steps for the prevention of such activities at workplace and put in place an appropriate mechanism of complaint/grievance redressal.
Management experience	The Project Proponent has the operational expertise, management strengths, financial capabilities and commitment to quality. They have expertise and experience in implementing similar project activities and engaging communities.
Gender equity in labor and work	The project abides the rules of equality accordingly and does not involve and is not complicit in any form of discrimination. Specific to gender equality, detailed enforcement rules are regulated in Women's Rights Protection Law to provide support and benefits to women. Qualified local residents, both men and women, are recruited to work for the project. During stakeholders' consultation process, comments were collected from the local people, including both men and women. The project abides the rules of equality. Regarding the gender equality, detailed enforcement rules are regulated in Labor Law and Women's Rights Protection Law.

Human trafficking, forced labor, and child labor	The Project will not use victims of human trafficking, forced labor, and child labor. Human trafficking is a serious crime that is strictly regulated and combated in the Indian legal system. India has promulgated the "Labor Law" to protect the legitimate rights and interests of workers and prohibit forced labor. India has published protection laws on the minors. Employment regulations are described in this Law. The Project requires skilled employees to operate, maintain, and manage the solar power plant. Therefore, the project does not employ and is not complicit in any form of child labor. According to the Employee list, it could be confirmed that the Project does not employ any child labor.

2.3.2 Human Rights

The PP emphasis on aligning the organizational practices with the human rights outlined in national and regional regulations. All customary rights to the land, waterbody and its resources are held by The New and Renewable Energy Department, Government of Madhya Pradesh (GoMP-NRE). The project respects protection of the rights of local communities and customary rights holders, aligning with applicable international human rights law, the United Nations Declaration.

2.3.3 Indigenous Peoples and Cultural Heritage

The project does not impact and protects cultural heritage as part of project activities.

2.3.4 Property Rights

Rights to territories and resources	The legal and customary rights of the land use and water body use during and post implementation of the project activity lies with The New and Renewable Energy Department, Government of Madhya Pradesh (GoMP-NRE). Project proponent have no rights on the legal ownership of the water body and land. There are no identified disputes or conflicts over rights of the water body and its resources. No concerns were received against the implementation of the project activity and confirms the consent and full support for the project implementation.

Respect for property rights	The project respects all stakeholder property rights. The Project Proponent will not alter or impact any of their rights, thus preserving and protecting them.

2.3.5 Benefit Sharing

Process used to design the benefit sharing plan	Not Applicable as there is no impact on the property rights of the Stakeholders due to the project activity. Hence, there is no benefit sharing involved.
Summary of the benefit sharing plan	Not Applicable as there is no impact on the property rights of the Stakeholders due to the project activity. Hence, there is no benefit sharing involved.
Approval and dissemination of benefit sharing plan	Not Applicable as there is no impact on the property rights of the Stakeholders due to the project activity. Hence, there is no benefit sharing involved.

2.4 Ecosystem Health

	Risks identified	Mitigation or preventative measure taken
Impacts on biodiversity and ecosystems	No risk identified	The project does not have any impact on biodiversity and ecosystems.
Soil degradation and soil erosion	No risk identified	Usage of land is very minimal in the project, as it is a floating solar project.
Water consumption and stress	No risk identified	The project involves water conservation measures to ensure the plant does not consume any fresh water.
Usage of fertilizers	No risk identified	There is no usage of fertilizers involved.

2.4.1 Rare, Threatened, and Endangered species

Is the project located in or adjacent to habitats for rare, threatened, or endangered species?

☐ Yes ☒ No

2.4.2 Introduction of species

N/A

2.4.3 Ecosystem conversion

As the project is not a ARR, ALM, WRC or ACoGS project, this section is not applicable.

3 APPLICATION OF METHODOLOGY

3.1 Title and Reference of Methodology

The project uses one large-scale CDM approved methodology.

Type (methodology, tool or module).	Reference ID	Title	Version
Methodology	ACM0002	ACM0002: Grid-connected electricity generation from renewable sources	21.0[2]
Tool	01	Tool for the demonstration and assessment of additionality	7.0.0[3]
Tool	05	Baseline, project and/or leakage emissions from electricity consumption and monitoring of electricity generation	3.0[4]
Tool	07	Tool to calculate the emission factor for an electricity system	7.0[5]
Tool	24	Common practice	3.1[6]
Tool	27	Investment analysis	13.0[7]

3.2 Applicability of Methodology

[2] https://cdm.unfccc.int/UserManagement/FileStorage/ZPFJL01OU2RYC6N3HASIXV7K84QBG9

[3] https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-01-v7.0.0.pdf

[4] https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-05-v3.0.pdf

[5] https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-07-v7.0.pdf

[6] https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-24-v1.pdf

[7] https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-27-v13.pdf

The applicability criteria of the applied methodology ACM0002: Grid-connected electricity generation from renewable sources, version21.0 are demonstrated below.

Methodology ID	Applicability condition	Justification of compliance
ACM0002	This methodology is applicable to grid-connected renewable power generation project activities that: (a) install Greenfield power plant; (b) involve a capacity addition to (an) existing plant(s); (c) involve a retrofit of (an) existing plant(s)/unit(s); (d) involve a rehabilitation of (an) existing plant(s)/unit(s); or (e) involve a replacement of (an) existing plant(s)/unit(s)	The project is a Greenfield (project is constructed at a site where no renewable energy power plant was operated prior to implementation of this project) grid connected renewable power plant. Therefore, it confirms to the said criteria (a).
	In case the project involves the integration of a BESS, the methodology is applicable to grid-connected renewable energy power generation project activities that: (a) Integrate BESS with a Greenfield power plant; (b) Integrate a BESS together with implementing a capacity addition to (an) existing solar photovoltaic or wind power plant(s)/unit(s); (c) Integrate a BESS to (an) existing solar photovoltaic or wind power plant(s)/unit(s) without implementing any other changes to the existing plant(s); (d) Integrate a BESS together with implementing a retrofit of (an) existing solar photovoltaic or wind power plant(s)/unit(s).	The project is the installation of a new grid connected renewable solar power project and does not involve the integration of a Battery Energy Storage System (BESS). This condition is not applicable for the project.
	The methodology is applicable under the following conditions: (a) Hydro power plant/unit with or without reservoir, wind power plant/unit, geothermal power plant/unit, solar power plant/unit, wave power plant/unit or tidal power plant/unit; (b) In the case of capacity additions, retrofits, rehabilitations or replacements (except for wind, solar, wave or tidal power capacity addition projects) the existing plant/unit started commercial operation prior to the start of a minimum historical reference period of five years, used for the	(a) The project is the installation of a Greenfield grid connected solar power plants with no BESS integration. Therefore, the methodology is applicable for the project. (b) The project is the installation of a Greenfield grid connected solar power plant without any capacity additions, retrofit, rehabilitations or replacements. (c) The project

	calculation of baseline emissions and defined in the baseline emission section, and no capacity expansion, retrofit, or rehabilitation of the plant/unit has been undertaken between the start of this minimum historical reference period and the implementation of the project; (c) In case of Greenfield project activities applicable under paragraph (a) above, the project participants shall demonstrate that the BESS was an integral part of the design of the renewable energy project (e.g. by referring to feasibility studies or investment decision documents); (d) The BESS should be charged with electricity generated from the associated renewable energy power plant(s). Only during exigencies may the BESS be charged with electricity from the grid or a fossil fuel electricity generator. In such cases, the corresponding GHG emissions shall be accounted for as project emissions following the requirements under section 5.4.4 below. The charging using the grid or using fossil fuel electricity generator should not amount to more than 2 per cent of the electricity generated by the project renewable energy plant during a monitoring period. During the time periods (e.g. week(s), months(s)) when the BESS consumes more than 2 per cent of the electricity for charging, the project participant shall not be entitled to issuance of the certified emission reductions for the concerned periods of the monitoring period.	is the installation of a Greenfield grid connected solar power plants with no BESS integration. (d) The project is the installation of a Greenfield grid connected solar power plants with no BESS integration. Therefore, this condition is not applicable.
	In case of hydro power plants, one of the following conditions shall apply: (a) The project is implemented in existing single or multiple reservoirs, with no change in the volume of any of the reservoirs; or (b) The project is implemented in existing single or multiple reservoirs, where the volume of the reservoir(s) is increased and the power density, calculated using equation (7), is greater than 4 W/m^2 ; or	The project activity is the installation of Greenfield grid-connected solar power plants/units. Therefore, the said criterion is not applicable.

	(c) The project results in new single or multiple reservoirs and the power density, calculated using equation (7), is greater than 4 W/m² ; or (d) The project is an integrated hydro power project involving multiple reservoirs, where the power density for any of the reservoirs, calculated using equation (7), is lower than or equal to 4 W/m², all of the following conditions shall apply: (i) The power density calculated using the total installed capacity of the integrated project, as per equation (8), is greater than 4 W/m² ; (ii) Water flow between reservoirs is not used by any other hydropower unit which is not a part of the project; (iii) Installed capacity of the power plant(s) with power density lower than or equal to 4 W/m² shall be: a. Lower than or equal to 15 MW; and b. Less than 10 per cent of the total installed capacity of integrated hydro power project.	
	In the case of integrated hydro power projects, project participants shall: (a) Demonstrate that water flow from upstream power plants/units spill directly to the downstream reservoir and that collectively constitute to the generation capacity of the integrated hydro power project; or (b) Provide an analysis of the water balance covering the water fed to power units, with all possible combinations of reservoirs and without the construction of reservoirs. The purpose of water balance is to demonstrate the requirement of specific combination of reservoirs constructed under CDM project for the optimization of power output. This demonstration has to be carried out in the specific scenario of water availability in different seasons to optimize the water flow at the inlet of power units. Therefore, this water balance will take into account seasonal flows from river, tributaries (if any), and rainfall for minimum of five years	The project activity is the installation of Greenfield grid-connected solar power plants/units. Therefore, the said criterion is not applicable.

	prior to the implementation of the CDM project.	
	The methodology is not applicable to: (a) Project activities that involve switching from fossil fuels to renewable energy sources at the site of the project, since in this case the baseline may be the continued use of fossil fuels at the site. (b) Biomass fired power plants/units;	(a) The project is the installation of greenfield solar power plant that does not involve fuel switch from fossil fuels to renewable energy. (b) The project is not a biomass fired power plant. Therefore, the methodology is applicable for this project.
	In the case of retrofits, rehabilitations, replacements, or capacity additions, this methodology is only applicable if the most plausible baseline scenario, as a result of the identification of baseline scenario, is "the continuation of the current situation, that is to use the power generation equipment that was already in use prior to the implementation of the project and undertaking business as usual maintenance".	The project is the installation of solar power plants/units without any retrofits, rehabilitations and replacements or capacity additions. Therefore, the said criterion is not applicable.
TOOL07	This tool may be applied to estimate the OM, BM and/or CM when calculating baseline emissions for a project that substitutes grid electricity that is where a project supplies electricity to a grid or a project that results in savings of electricity that would have been provided by the grid (e.g., demand-side energy efficiency projects).	The project is a Greenfield solar power generation plant and hence, according to the applied methodology, the baseline scenario is electricity delivered to the grid by the project would have otherwise been generated by the operation of grid-connected power plants and by the addition of new generation sources, as reflected in the combined margin (CM) calculations described in "Tool 07: Tool to calculate the emission factor

		for an electricity system", version 7.0.
	Under this tool, the emission factor for the project electricity system can be calculated either for grid power plants only or, as an option, can include off-grid power plants. In the latter case, two Sub-options under the step 2 of the tool are available to the project participants, i.e., option IIa and option IIb. If option IIa is chosen, the conditions specified in "Appendix 2: Procedures related to off-grid power generation" should be met. Namely, the total capacity of off-grid power plants (in MW) should be at least 10 per cent of the total capacity of grid power plants in the electricity system; or the total electricity generation by off-grid power plants (in MWh) should be at least 10 per cent of the total electricity generation by grid power plants in the electricity system; and that factors which negatively affect the reliability and stability of the grid are primarily due to constraints in generation and not to other aspects such as transmission capacity.	Since the project is grid connected, this condition is applicable and the emission factor has been calculated accordingly. Combined margin grid emission factor has been calculated as per the CO_2 emission factor database published by the CEA in which for the calculation of emission factor CEA have only considered grid connected plants.
	In case of CDM projects the tool is not applicable if the project electricity system is located partially or totally in an Annex I country.	The project activity is located in India, a non-Annex I country. Therefore, this tool is applicable for the project activity
	Under this tool, the value applied to the CO_2 emission factor of bio fuels is zero	The project is a grid connected solar power project and therefore, this criterion is not applicable for the project
TOOL01	The use of the "Tool for the demonstration and assessment of additionality" is not mandatory for project participants when proposing new methodologies. Project	The project owner is not proposing any new methodology and applied this tool for the demonstration of additionality in compliance

	participants may propose alternative methods to demonstrate additionality for consideration by the Executive Board. They may also submit revisions to approved methodologies using the additionality tool.	with the tool. Refer to section 3.5 of the PD for the detailed applicability of this tool and additionality assessment. Hence this tool is applicable
	Once the additionally tool is included in an approved methodology, its application by project participants using this methodology is mandatory.	In line with the methodology requirement, Project owner has applied the TOOL01: Tool for the demonstration and assessment of additionality, Version 7.0.0. for the demonstration of additionality. Hence, this tool is applicable.
	The use of the "Tool for the demonstration and assessment of additionality" is not mandatory for project participants when proposing new methodologies. Project participants may propose alternative methods to demonstrate additionality for consideration by the Executive Board. They may also submit revisions to approved methodologies using the additionality tool.	The project owner is not proposing any new methodology and applied this tool for the demonstration of additionality in compliance with the tool. Refer to section 3.5 of the PD for the detailed applicability of this tool and additionality assessment. Hence this tool is applicable
TOOL24	This methodological tool is applicable to project activities that apply the methodological tool "Tool for the demonstration and assessment of additionality", the methodological tool "Combined tool to identify the baseline scenario and demonstrate additionality", or baseline and monitoring methodologies that use the common practice test for the demonstration of additionality.	Project applies "Tool 01: Tool for the demonstration and assessment of additionality", version 7.0.0. Hence, this tool is applicable.
	In case the applied approved baseline and monitoring methodology defines approaches for the conduction of the common practice test that are different from those described in this methodological tool, the requirements contained in the methodology shall prevail.	Applied methodology ACM0002 "Grid-connected electricity generation from renewable sources", version 21.0 doesn't specify any approach for the demonstration of common practice analysis. As per the

		methodology the additionality including common practice analysis has been demonstrated as per the Tool 01: "Tool for the demonstration and assessment of additionality" version 7.0.0 and Tool 24: Common Practice, version 3.1. Hence Justified.
TOOL27	This methodological tool is applicable to project activities that apply the methodological tool "Tool for the demonstration and assessment of additionality", the methodological tool "Combined tool to identify the baseline scenario and demonstrate additionality", the guidelines "Non-binding best practice examples to demonstrate additionality for SSC project activities", or baseline and monitoring methodologies that use the investment analysis for the demonstration of additionality and/or the identification of the baseline scenario.	Project applies Tool 01 "Tool for the demonstration and assessment of additionality", version 7.0.0. Hence this tool is applicable.
	In case the applied approved baseline and monitoring methodology contains requirements for the investment analysis that are different from those described in this methodological tool, the requirements contained in the methodology shall prevail.	As per the methodology the additionality including investment analysis has been demonstrated as per the Tool 01: "Tool for the demonstration and assessment of additionality" version 7.0.0 and Tool 27: Investment Analysis version 13.0 Hence Justified.
TOOL05	Paragraph 05 If emissions are calculated for electricity consumption, the tool is only applicable if one out of the following three scenarios applies to the sources of electricity consumption:	The project activity is the Greenfield grid connected solar power project. The project activity consumes electricity from the grid and the emissions for the electricity consumption from

	(a) Scenario A: Electricity consumption from the grid. The electricity is purchased from the grid only, and either no captive power plant(s) is/are installed at the site of electricity consumption or, if any captive power plant exists on site, it is either not operating or it is not physically able to provide electricity to the electricity consumer; (b) Scenario B: Electricity consumption from (an) off-grid fossil fuel fired captive power plant(s). One or more fossil fuel fired captive power plants are installed at the site of the electricity consumer and supply the consumer with electricity. The captive power plant(s) is/are not connected to the electricity grid; or Scenario C: Electricity consumption from the grid and (a) fossil fuel fired captive power plant(s). One or more fossil fuel fired captive power plants operate at the site of the electricity consumer. The captive power plant(s) can provide electricity to the electricity consumer. The captive power plant(s) is/are also connected to the electricity grid. Hence, the electricity consumer can be provided with electricity from the captive power plant(s) and the grid.	the grid is calculated. Hence, the scenario A is applicable.
	Paragraph 06 This tool can be referred to in methodologies to provide procedures to monitor amount of electricity generated in the project scenario, only if one out of the following three project scenarios applies to the recipient of the electricity generated: (a) Scenario I: Electricity is supplied to the grid; (b) Scenario II: Electricity is supplied to consumers/electricity consuming facilities; or	This tool is referred in line with the methodology “ACM0002: Grid-connected electricity generation from renewable sources”, Version 21.0. Scenario I is applicable for the project activity as the electricity generated from the grid is being supplied to the grid.

	(c) Scenario III: Electricity is supplied to the grid and consumers/electricity consuming facilities.	
	Paragraph 07 This tool is not applicable in cases where captive renewable power generation technologies are installed to provide electricity in the project, in the baseline scenario or to sources of leakage. The tool only accounts for CO_2 emissions.	This is not a captive power project where the generated power is used for captive consumption. The solar power project is installed to supply the generated electricity to the national grid through the Power purchase agreement with DISCOM. Hence, this condition is not applicable.

3.3 Project Boundary

The project boundary includes the solar project, sub-stations, Indian grid and all power plants connected to the grid. The project activity will evacuate power to the Indian grid. Therefore, the entire Indian grid and all grid connected power plants have been considered in the project boundary for the project activity.

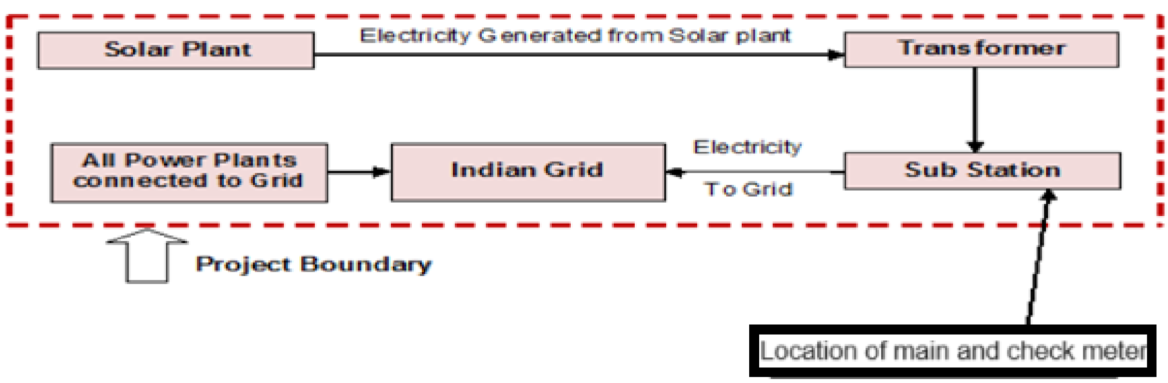


Figure: Project Boundary

The project does not involve any other emissions sources not foreseen by the methodologies. The greenhouse gases and emission sources included in or excluded from the project boundary are shown in table below.

Source		Gas	Included?	Justification/Explanation
Baseline	Grid Connected Electricity Generation	CO_2	Yes	Main Emission Source
		CH_4	No	Minor Emission source
		N_2O	No	Minor Emission source

Source		Gas	Included?	Justification/Explanation
		Other	-	-
Project	Greenfield Solar PV Power Project activity	CO_2	No	Project activity does not emit CO_2
		CH_4	No	Project activity does not emit CH_4
		N_2O	No	Project activity does not emit N_2O
		Other	-	-

3.4 Baseline Scenario

An approved large-scale baseline CDM methodology ACM0002 "Grid-connected electricity generation from renewable sources", Version 21.0, has been followed along with the Tool 07 "Tool to calculate the emission factor for an electricity system, version 7.0" is used to establish the baseline scenario.

According to the methodology baseline scenario has been identified as "Electricity delivered to the grid by the project activity would have otherwise been generated by the operation of grid-connected power plants and by the addition of new generation sources, as reflected in the combined margin (CM) calculations described in the Tool 07 "Tool to calculate the emission factor for an electricity system", version 7.0 means that a power plant with emission factor equivalent to grid mix would have supplied electricity in the absence of new project plant or added capacity. A grid emission factor is a reasonable benchmark that provides the proxy performance of the baseline power plant.

The project involved setting up of floating solar power generation plant to harness the power of solar energy to produce electricity and supply to the grid. In the absence of the project activity, the equivalent amount of power would have been supplied to the electricity grid by the operation of grid-connected power plants and by the addition of new generation sources, as reflected in the combined margin (CM) calculations.

Hence, the baseline for the project activity is the equivalent amount of power from the Indian grid.

During the implementation of the project activity and establishing the baseline scenario the relevant national and/or sectoral policies, regulations and circumstances are taken into account.

- Implementation of solar energy-based power projects for electricity generation is not mandatory under any law in India, the project activity is a voluntary action.
- Despite the gradual increase in renewable energy sources (including solar energy) in power sector, still about two-third of installed power generation capacity is based on fossil fuel-based energy sources, hence the national grid is fed by electricity generated predominantly by the fossil-fuel based power plants.

There is no legal and regulatory requirement that mandates the production of energy by the chosen technology. Investment in solar energy projects in the State of Madhya Pradesh and the Indian electricity grid are not mandatory. There are no national or local laws or regulations that require this investment to be undertaken, i.e., setting up of solar power projects. The setting up of solar energy projects is a voluntary activity.

In the absence of the project activity, the equivalent amount of electricity would have been drawn from the Indian grid. The combined margin ($EF_{grid,CM,y}$) is the result of a weighted average of two emission factor pertaining to the electricity system: the operating margin (OM) and build margin (BM). Calculations for this combined margin must be based on data from an official source (where available) and made publicly available. The CEA database version 19.0 is the latest available data at the time of initial submission, hence same is considered for emission factor

The combined margin emission factor of the Indian grid used for the project activity is as follows

Parameter	Value	Nomenclature	Source
$EF_{grid,CM,y}$	0.9352 tCO_2/MWh	Combined margin CO_2 emission factor for the project electricity system in year y	Calculated as the weighted average of the operating margin (0.75) & build margin (0.25) values, sourced from Baseline CO_2 Emission Database, Version 19.0[8] published by Central Electricity Authority (CEA), Government of India
$EF_{grid,OM,y}$	0.9580 tCO_2/MWh	Operating margin CO_2 emission factor for the project electricity system in year y	Calculated as the last 3 years (2020-21 ,2021-22& 2022-23,) generation-weighted average, sourced from Baseline CO_2 Emission Database, Version 19.0, published by Central Electricity Authority (CEA), Government of India
$EF_{grid,BM,y}$	0.8670 tCO_2/MWh	Build margin CO_2 emission factor for the project electricity system in year y	Baseline CO_2 Emission Database, Version 19.0, published by Central Electricity Authority (CEA), Government of India

The baseline case is in compliance with all applicable legal and regulatory requirements references.

3.5 Additionality

The additionality of the project activity is demonstrated by following the guidance provided in the VCS project standard Version 4.6.

As per section 3.14 in VCS Standard v4.6, additionality can be demonstrated using the following components.

3.5.1 Regulatory Surplus

[8] https://cea.nic.in/wp-content/uploads/2021/03/Baseline_Carbon_Dioxide_Emission_Database_Version_19.0.zip

Is the project located in an UNFCCC Annex 1 or Non-Annex 1 country?

☐ Annex 1 country ☒ non-annex 1 country

Are the project activities mandated by any law, statute, or other regulatory framework?

☐ Yes ☒ No

If the project is located inside a non-annex 1 country and the project activities are mandated by a law, statute, or other regulatory framework, are such laws, statutes, or regulatory frameworks systematically enforced?

☐ Yes ☒ No

As per 3.14.1 demonstrating regulatory surplus, the Project conforms to all the applicable laws and regulations in India:

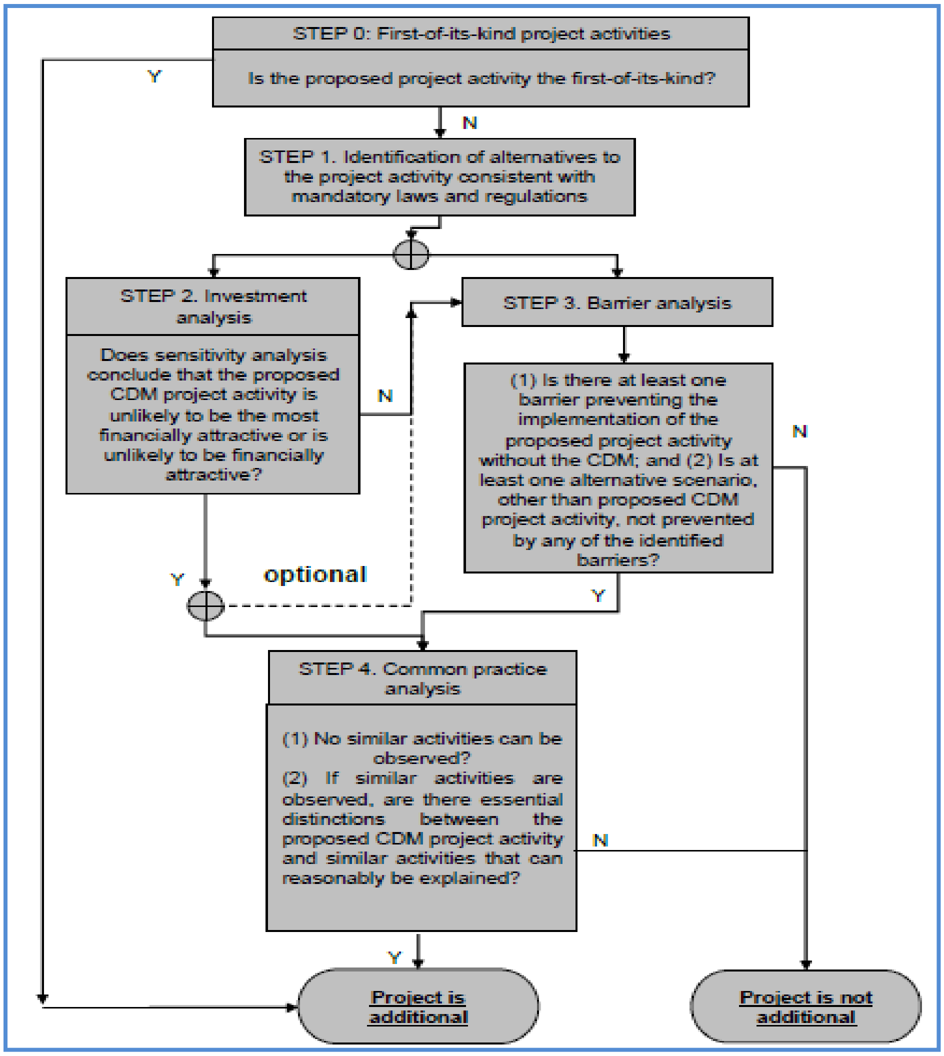
- Power generation using renewable energy is not a legal requirement or a mandatory option.
- The Indian Electricity Act, 2003 (May 2007 Amendment) does not influence the choice of fuel used for power generation.
- There is no legal requirement on the choice of a particular technology for power generation
- Both the baseline and project activity are in compliance with laws and regulations required. There is no mandatory requirement to implement the project activity.
- The project activity comes under white category as per local regulation, thus there shall be no necessity of obtaining the Consent to Operate for White category of industries. Since project activity falls under white category and the non-polluting nature of project fulfils the compliance to the local laws and regulations.

As per the section 3.14 of the VCS standard version 4.6, the project is not implemented by the force of law. This is a voluntary activity undertaken by the project owner in compliance with all the legal requirement in the host country. Hence project demonstrates regulatory surplus.

3.5.2 Additionality Methods

Additionality has been demonstrated as per the applied methodology ACM0002 "Grid-connected electricity generation from renewable sources" (Version 21.0). Methodology requires the project owner to determine the additionality based on TOOL01 "Tool for the demonstration and assessment of additionality", Version 7.0.0.

The stepwise approach to establish additionality of the project activity has been followed, details of which are provided in the following paragraphs:



As per the applied methodology requirement, Additionality of the project activity is demonstrated using the Tool 01 "Tool for the demonstration and assessment of additionality" Version 7.0.0. The tool defines the following steps:

Sub Step 0: Demonstration whether the proposed project activity is the first-of-its-kind.

The project activity is not the first of its kind as implementation of floating solar power project in the country is not first of its kind.

Step 1: Identification of alternatives to the project activity consistent with current laws and regulations

<u>Sub-step 1a: Define alternatives to the project activity</u>

As per the applied ACM0002 "Grid-connected electricity generation from renewable sources", version 21.0; Para 24, if the project activity is the installation of a Greenfield power plant with or without a BESS, the baseline scenario is electricity delivered to the grid by the project activity would have otherwise been generated by the operation of grid connected power plant and by the addition of new generation sources.

However, for the assessment of additionality the following alternatives are identified:

Alternative 1: The proposed project activity undertaken without being registered as a VCS project activity.

The Project Owner could proceed with the implementation of the project without considering the carbon credit benefits. The electricity produced from the renewable energy project would have been sold to the grid. This is in compliance with all applicable legal and regulatory requirements and can be a part of the baseline. However, the Project activity is not feasible without revenues from sale of Carbon Credits. This argument has been discussed in step 2 of the Additionality section.

Alternative 2: No project activity is undertaken and equivalent amount of energy would have been produced by the grid electricity system through its currently running power plants and by new capacity addition to the grid i.e., Continuation of the present situation.

The project owner would have continued without investment in Project activity with usual business activities.

The electricity delivered to the grid by the project activity would be generated by the operation of grid-connected power plants and by the addition of new generation sources into the grid.

It means that a power plant with emission factor equivalent to grid mix would have supplied electricity in absence of new project plant or added capacity. A grid emission factor is a reasonable benchmark that provides the proxy performance of the baseline power plant.

Outcome of Sub-step 1a:

All the realistic alternatives for the project activity have been enlisted above.

Thus, though two alternatives are mentioned above as per step of additionality tool, the first alternative is not possible as project activity is not viable without carbon credit benefits and second alternative is the baseline scenario for the project activity as per methodology as mentioned in section 3.4 of PD.

It is to be noted that being the greenfield project activity, "the baseline scenario is electricity delivered to the grid by the project activity would have otherwise been generated by the operation of grid-connected power plants and by the addition of new generation sources, as reflected in the combined margin (CM) calculations described in the Tool 07 "Tool to calculate the emission factor for an electricity system" Version 7.0.

Sub-step 1b: Consistency with mandatory laws and regulations:

The alternative(s) shall be in compliance with all applicable legal and regulatory requirements, even if these laws and regulations have objectives other than GHG reductions, e.g., to mitigate local air pollution. The project activity comes under white category as per local regulation, thus there shall be no necessity of obtaining the Consent to Operate'' for White category of industries. Since project activity falls under white category and the non-polluting nature of project fulfils the compliance to the local laws and regulations (This sub-step does not consider national and local policies that do not have legally-binding status).

The relevant national laws and regulations pertaining to generation of energy in India are:

- Electricity Act, 2003
- National Electricity Policy, 2005

- The Electricity (Supply) Act, 1948
- The Electricity Regulation Commission Act, 1998
- Tariff Policy, 2006
- The factories act, 1948
- Schedule 1 of Ministry of Environment and Forest notification

The Project activity conforms to all the applicable laws and regulations in India:

- Power generation using renewable energy is not a legal requirement or a mandatory option.
- There are state and sectoral policies, framed primarily to encourage solar power projects.
- These policies have also been drafted realizing the extent of risks involved in the projects and to attract private investments.
- The Indian Electricity Act, 2003 (May 2007 Amendment) does not influence the choice of fuel used for power generation.
- There is no legal requirement on the choice of a particular technology for power generation.

The both alternatives are in compliance with laws and regulations required. There is no mandatory requirement to implement the project activity.

Outcome of Sub-step 1b:

Hence, both the alternatives enlisted above are found to comply with the mandatory laws and regulations taking into account the enforcement of the legislations in the region or country and EB decisions on national and/or sectoral policies and regulations. Since solar projects are categorized as white category, no consent to operate required from pollution control board.

However, Alternative 2 has been selected as the appropriate baseline alternative for this project activity in line with methodology.

Step 2: Investment Analysis

The investment analysis has been done in order to make an economic and financial evaluation of the project. To conduct the investment analysis, Methodological tool: Tool 27: “Investment analysis”, version 13.0, has been referred. No public funding or ODA are available in India for finance of this type of projects. For investment analysis, financial assumptions from the debt, finance from the lenders have been considered from the standard market rates available for the similar projects and enquiries from the vendors.

As per para 29 of Tool 01 “Tool for the demonstration and assessment of additionality”, version 7.0.0. It is determined that the project activity is not an economically attractive or financially feasible option.

To conduct the investment analysis, Methodological Tool 27: “Investment analysis”, version 13.0, EB 120 Annex 3 has been referred.

Sub-step 2a: Determine appropriate analysis method

As per Tool 01 “Tool for the demonstration and assessment of additionality” (version 7.0.0), for financial analysis of the project, the following three options are available:

Option I: Simple Cost Analysis

Option II: Investment Comparison Analysis

Option III: Benchmark Analysis

The project will generate revenues from sale of electricity; therefore, Option I is not applicable in line with para 32 of the Methodological tool: “Tool for the demonstration and assessment of additionality”, version 7.0.0. Same applies to the Option II which is applied in case there are alternatives to the project activity as per para 42 of the “Tool for the demonstration and assessment of additionality”, version 7.0.0.

Since, identified baseline for the project activity is continuation of current practice (i.e. equivalent amount of energy would have been generated by grid electricity system through its currently operating power plants and by new capacity addition) and which is outside the direct control of the project owner, hence benchmark analysis (option III), where the returns on investment in the project activity are compared to benchmark returns that are available to any investors in the country is selected as the most appropriate method.

Sub-step 2b: Option III. Apply benchmark analysis

As per para 15 of EB 120 Annex 3 states that required/expected returns on equity are appropriate benchmarks for equity IRR. The project owner has chosen benchmark analysis to demonstrate the additionality of the project. The project is promoted by private limited company and hence the return on equity and the risks associated with the investments for their shareholder is of primary concern. Hence, in order to analyze the financial viability of the project activity, the prime financial indicator that has been used is the post-tax equity IRR of the project activity.

As per Investment Analysis tool, required/expected returns on equity are appropriate benchmarks for a post-tax equity IRR. The post-tax equity IRR is considered as the financial indicator and the benchmarks used is cost of equity. Hence the benchmarks used are applicable to the project activity and the type of IRR calculation presented.

Selection of Appropriate Benchmark

The benchmark has been considered in accordance with Guidance of Tool 27: Investment Analysis, latest version available at the time of initial submission.

Default value in the Tool 27: Investment analysis, version 13.0:

The default value as per latest version of Tool 27: Investment analysis, version 13.0 is 9.13% for group 1 project in India is used which is appropriate for benchmark calculation and the project owner has considered the same tool for default value of return on equity for the respective project.

Table under Appendix in EB120, Annex 3 specifies default value of expected return on equity in real terms for Energy Industries (Group 1) in India = **9.13%**[9]

Thus, minimum cost of equity considered for calculation of Benchmark = 9.13%

[9] https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-27-v13.pdf

The Required return on equity (benchmark) was computed in the following manner:

Nominal Benchmark[10] = {(1+Real Benchmark) * (1+Inflation rate)} -1

Where:

Default value for Real Benchmark = 9.13% (as per Appendix of EB 120, Annex 3)

Thus, the minimum cost of equity considered for calculation of Benchmark = 9.13%

Inflation Rate:

As per paragraph 16 of Appendix A of the above-mentioned document, "In situations where an investment analysis is carried out in nominal terms, project owners can convert the real term values provided in the table to nominal values by adding the inflation rate. The inflation rate shall be obtained from the inflation forecast of the central bank of the host country for the duration of the crediting period. If this information is not available, the target inflation rate of the central bank shall be used. If this information is also not available, then the average forecasted inflation rate for the host country published by the IMF (International Monetary Fund World Economic Outlook) or the World Bank for the next five years after the start of the project activity shall be used".

In line with investment analysis tool, Project owner has considered the next 5 years forecasted inflation rate published by the International Monetary Fund World Economic Outlook[11] at the time of project investment decision (i.e., SJVN Pre-Bid Note) i.e., 22/04/2022. The applicable inflation rate and corresponding benchmark values are provided below for both the projects.

Inflation Forecast	Benchmark
4.32%	13.84%

The benchmark of **13.84 %** has been considered for the investment analysis.

Sub-step 2c: Calculation and comparison of financial indicators

The period considered for Post Tax Equity IRR calculations is 25 years, which corresponds to the operational lifetime of the project activity.

Depreciation, and other non-cash items related to the project activity, which have been deducted in estimating gross profits on which tax is calculated, is added back to net profits for the purpose of calculating the financial indicator.

Chronology of events of the project activity are provided below:

90 MW Floating Solar Project - Omkareshwar, Madhya Pradesh		
S. No.	**Document**	**Chronology of the project**
1	RfP date	02-11-2021
2	SJVN Pre-Bid Note	22-04-2022
3	Last date for Bid Submission	26-04-2022
4	LoA to SJVN	30-06-2022

[10] As per Pg. 320 of Corporate Finance, Second Edition of Aswath Damodaran
[11] https://www.imf.org/en/Publications/WEO/weo-database/2022/October/weo-report?c=534,&s=PCPI,PCPIPCH,&sy=2022&ey=2027&ssm=0&scsm=1&scc=0&ssd=1&ssc=0&sic=0&sort=country&ds=.&br=1

5	Power Purchase Agreement (PPA)	04-08-2022
6	Implementation Support Agreement	04-08-2022
7	Land Use Permission Agreement	04-08-2022
8	Signing of EPC Contract	29-12-2022
9	Commissioning Date (Expected)	12-04-2023

The investment decision date of the project is 22/04/2022, which is the date when SJVN signed the financial parameters for e-bidding process with committees' approval. Input values used in the investment analysis are sourced from the Pre-Bid note which is available at the time of the investment decision. The estimated PLF has been sourced from the "Energy Yield Assessment Report" prepared by third party engineering company which is also available at the time of investment decision making which is also in line with the guidelines of EB48 Annex 11.

Input values considered for the IRR calculation are provided below.

Particulars	Value	Unit	Source/Remarks
Capacity of the project	90	MW	Pre-Bid Note
Annual Net generation (First year)	180.63	GWh	PVsyst Report
Plant Load Factor	22.91%	%	Calculated
Annual Degradation	0.5%	%	Pre-Bid Note
Project cost	5971.00	INR Million	Pre-Bid Note
Debt	80%	%	Pre-Bid Note
Equity	20%	%	
Debt	4776.80	INR Million	Calculated
Equity	1194.20	INR Million	Calculated
Interest rate	6.00%	%	Pre-Bid Note
Debt Repayment tenure	20	years	Pre-Bid Note
Moratorium	0	year	
Operation and Maintenance	45.7	INR Million	Pre-Bid Note
Escalation in O & M	3.84%	%	Pre-Bid Note
Insurance & overhead	8.96	INR Million / Yr	Internal assumption
Tariff	3.72	Rs/kWh	Pre-Bid Note
Gross Depreciable Value	5971.00	INR Million	Calculated
Salvage Value (@ 10%)	597.1	INR Million	Calculated
Net Depreciable Value	5373.90	INR Million	Calculated
Residual Value	597.10	INR Million	Calculated
Depreciation Rate (Book)	4.00%	%	Calculated
IT Depreciation Rate	7.69%	%	As Per Income Tax, Depreciation rates for power generating units http://www.incometaxindia.gov.in/charts%20%20tables/depreciation%20rates.htm
Income tax rate	34.94%	%	Calculated
MAT rate	17.47%	%	Calculated
Salvage Value	10%	%	

Project lifetime	25	years	

Tax Rates

Financial Year	FY 2022-23		
Income tax rate (%)	30.00%	Indian IT Act for FY 2022-23	https://www.incometax.gov.in/iec/foportal/help/company/return-applicable
MAT (%)	15.00%	Indian IT Act for FY 2022-23	
Surcharge (%)	12.00%	Indian IT Act for FY 2022-23	
Education cess (%)	4.00%	Indian IT Act for FY 2022-23	

Final Tax Rates

Income tax rate (%)	34.94%	%	Calculated
MAT (%)	17.47%	%	Calculated

Post Tax Equity IRR for all the project activities against the benchmark values are shown in table below. Thus, it is evident that the project is not financially attractive as the equity IRR is less below the benchmark value.

Post tax Equity IRR	Benchmark Value
9.63%	13.84%

The robustness of the conclusion drawn above, namely that the project is not financially attractive, has been tested by subjecting critical assumptions to reasonable variation. As required by Annex 2 of EB 120, only variables, including the initial investment cost, that constitute more than 20% of either total project costs or total project revenues should be subjected to reasonable variation. Project owner has identified the total revenue from the project activity is dependent on the Tariff, Net Annual Generation, Project Cost and O&M Costs constitute more than 20% of the project costs. These factors have been subjected to a 10% variation on either side and the results of the sensitivity analysis indicate that even after applying such variation the EIRR does not cross the benchmark.

Variation %	-10%	Normal	10%	Variation required to breach benchmark	Value required to breach benchmark
Tariff (INR/kWh)	6.14%	9.63%	13.04%	12.30%	4.17
Net Annual Generation (GWh)	6.14%	9.63%	13.04%	12.30%	202.85
Project Cost (INR Mn)	13.07%	9.63%	6.75%	-11.90%	5260.45
O&M Cost (INR Mn)	9.85%	9.63%	9.28%	-138.00%	-17.36

An analysis has been done to identify the percentage variation at which the financial indicators will equal/breach the benchmark and the probability of its occurrence. Based on sensitivity analysis it can be concluded that the project activity is additional with reasonable variation in values and is not likely to reach the benchmark value. The occurrence of these events is unlikely for the following reasons:

a) **Tariff:** The Tariff rate of electricity used for investment analysis i.e., 3.72 INR/kWh is sourced from the Pre-Bid note available at the time of investment decision. Furthermore, the project will breach the benchmark value at a tariff variation of 12.30%. The power purchase agreement is signed for 25 years at fixed rate of 3.26 INR/kWh which is less than the estimated tariff, an increase in tariff is unlikely.

b) **Net Annual Generation:** The Net Annual Generation considered is based on the Third-Party report i.e., 180.63 GWh and the Net Annual Generation breach the benchmark value at variation of more than 12.30%. The chances of increase in generation value to breach the benchmark is very low as the considered generation is based on the energy yield assessment study.

c) **Project Cost:** The project cost considered for investment analysis i.e., 5971 million INR. The cost is sourced from financial parameters provided by SJVN which is based on the negotiations with Supplier. A variation of -11.90% is required for IRR to breach benchmark.

d) **O&M Costs:** The Operation and Maintenance cost is considered from the SJVN Financial Parameters applicable at the time of investment decision. Sensitivity analysis reveals that O&M will breach the benchmark at negative values and is hypothetical case. Since the O&M cost is subject to escalation and subject to inflationary pressure, any reduction in the O&M costs is highly unlikely.

Step 3: Barrier analysis

Barrier analysis has not been used.

Step 4: Common practice analysis

SJVN Ltd. as a parent company formed a SPV (Special purpose vehicle) in the name of SJVN Green Energy Limited (SGEL) for the implementation of solar projects. Thus, common practice analysis has been carried out for the large scale bundled project activity.

Stepwise approach for common practice analysis has been carried out as per Methodological Tool 24 “Common Practice”, version 03.1 EB84, Annex 7:

(a) The projects are located in the applicable geographical area;
(b) The projects apply the same measure as the proposed project activity;
(c) The projects use the same energy source/fuel and feedstock as the proposed project activity, if a technology switch measure is implemented by the proposed project activity;
(d) The plants in which the projects are implemented produce goods or services with comparable quality, properties and applications areas (e.g., clinker) as the proposed project plant;
(e) The capacity or output of the projects is within the applicable capacity or output range calculated in Step 1;
(f) The projects started commercial operation before the project design document (CDM-PDD) is published for global stakeholder consultation or before the start date of proposed project activity, whichever is earlier for the proposed project activity.

Steps Involved & Outputs

Step (1): *Calculate applicable capacity or output range as +/- 50% of the total design capacity or output of the project activity:*

The capacity of the project activity is 90 MW and hence the output range as per the guideline is selected to be 45 MW to 135 MW.

Step (2): *Identification of the similar projects (CDM and non-CDM) is carried out as per sub-steps of Step (2) as follows:*

a) The project is located in Madhya Pradesh state of India. The project has been awarded through the RfP released by RUMSL where the project activity can be located only in the Omkareshwar dam of Madhya Pradesh. Therefore, projects located in Madhya Pradesh state have been chosen for analysis.
b) The projects applying same measure (i.e., only renewable energy through floating Solar) are selected, as the project is floating solar power project.

   Therefore, all projects applying same measure (b) as the project activity are candidates for similar projects.
c) The energy source used by the project activity is Solar. Hence, only solar energy projects have been considered for analysis.
d) The project activity produces electricity; therefore, all power plants that produce electricity are candidates for similar projects.
e) The capacity range of the projects is within the applicable capacity range for the bundled project is 45 MW to 135 MW
f) As per the tool 24: Common practice, version 3.1 and the CDM definitions, the start date of the project is 29/12/2022(Signing of the EPC Contract). As Kyoto Protocol was ratified by India on 26/08/2002[12], therefore projects which had started commercial operation between 26/08/2002 to 29/12/2022 have been identified for common practice analysis.

Numbers of Similar projects identified which fulfill above-mentioned conditions are zero in the selected geographical area. Therefore,

$N_{solar} = 0$

Step (3): *within the projects identified in Step 2, identify those that are neither registered CDM project activities, project activities submitted for registration, nor project activities undergoing validation. Note their number, N_{all}.*

From the projects identified above, Numbers of projects that are neither registered in CDM project activities, project activities submitted for registration, nor project activities undergoing validation is Zero. Therefore,

$N_{all} = 0$

Step (4): *within similar projects identified in Step 3, identify those that apply technologies that are different to the technology applied in the proposed project activity. Note their number N_{diff}.*

[12] https://unfccc.int/node/61082

Since the projects identified above is Zero, and the number of projects that apply technologies that are different to the technology applied in the proposed project activity has been identified as Ndiff.

N_{diff} = 0

Step (5): *calculate factor $F=1-N_{diff}/N_{all}$ representing the share of similar projects (penetration rate of the measure/technology) using a measure/technology similar to the measure/technology used in the proposed project activity that deliver the same output or capacity as the proposed project activity.*

Calculate **$F = 1-N_{diff}/N_{all}$**

F = 1-(0/0) = **1**

Nall-Ndiff =0

As per Tool 24 "Common practice" version 03.1, the project activity is a "Common practice" within a sector in the applicable geographical area if the factor F is greater than 0.2 and N_{all} – N_{diff} is greater than 3. Thus, if both conditions are fulfilled, then project activity will be a common practice. Otherwise, the project activity is treated as not a common practice.

Outcome of Step 5:

As,

i. F = 1; which is greater than 0.2
ii. N_{all} – N_{diff} =0; which is not greater than 3

The project activity does not satisfy second condition. Hence, project activity is not a common practice.

Thus, the project activity is not a "common practice" within a sector in the applicable geographical area.

Conclusion:

The Project fulfils all necessary requirements of additionality specified in the Tool 01: "Tool for the demonstration and assessment of additionality", Version 7.0.0. Hence, the project is additional.

3.6 Methodology Deviations

This section is not applicable, as there is no deviation of methodology applied by the Project.

4 QUANTIFICATION OF ESTIMATED GHG EMISSION REDUCTIONS AND REMOVALS

4.1 Baseline Emissions

As per the approved consolidated Methodology ACM0002 "Grid-connected electricity generation from renewable sources" version 21.0, the Baseline emissions include only CO_2 emissions from electricity generation in grid-connected power plants that are displaced due to the project activity. The methodology assumes that all project electricity generation above baseline levels would have been generated by existing grid-connected power plants and the addition of new grid- connected power plants. The baseline emissions are to be calculated as follows:

$$BE_y = EG_{PJ,y} \times EF_{grid,CM,y}$$

Where,

BE_y = Baseline emissions in year y (tCO_2/yr)

$EG_{PJ,y}$ = Quantity of net electricity generation that is produced and fed into the grid as a result of the implementation of the CDM project activity in year y (MWh/yr)

$EF_{grid,CM,y}$ = Combined margin CO_2 emission factor for grid connected power generation in year y calculated using the latest version of the "Tool to calculate the emission factor for an electricity system" (tCO_2/MWh)

As per para 49 of ACM0002, "Grid-connected electricity generation from renewable sources" version 21.0, when the project activity is installation of Greenfield Power Plant, then:

$$EG_{PJ,y} = EG_{facility,\ y}$$

Where,

$EG_{PJ,y}$ = Quantity of net electricity generation that is produced and fed into the grid as a result of the implementation of the CDM project activity in year y (MWh/yr)

$EG_{facility,\ y}$ = Quantity of net electricity generation supplied by the project plant/unit to the grid in year y (MWh/yr)

The $EG_{facility,y}$ is estimated for entire crediting period of the Project considering the annual degradation of 0.50% in the generation and PLF provided as per the third-party engineering company report as below:

ACM0002 Version 21.0	
Crediting Period Year	Total Net Electricity generation
12-Apr-2024 to 31-Dec-2024	130,641
01-Jan-2025 to 31-Dec-2025	179,719
01-Jan-2026 to 31-Dec-2026	178,820

01-Jan-2027 to 31-Dec-2027	177,926
01-Jan-2028 to 31-Dec-2028	177,037
01-Jan-2029 to 31-Dec-2029	176,151
01-Jan-2030 to 31-Dec-2030	175,271
01-Jan-2031 to 11-Apr-2031	48,257
Total	**1,243,822**

$EG_{facility, y}$ (MWh)
1,243,822

Calculating the Emission Factor ($EF_{grid,CM,y}$)

As per the methodology combined margin grid emission factor has been calculated as per the Tool 07 **"Tool to calculate the grid emission factor for an Electricity System"** version 7.0

CO_2 Baseline Database for the Indian Power Sector, Version 19, November 2023[13] published by Central Electricity Authority (CEA), Government of India has been used for the calculation of emission reduction.

As per the Tool 07 "Tool to calculate the emission factor for an electricity system" Version 7.0, the following steps have been followed.

(a)**Step 1:** Identify the relevant electricity systems;
(b)**Step 2:** Choose whether to include off-grid power plants in the project electricity system (optional);
(c)**Step 3:** Select a method to determine the operating margin (OM);
(d)**Step 4:** Calculate the operating margin emission factor according to the selected method;
(e)**Step 5:** Calculate the build margin (BM) emission factor;
(f)**Step 6:** Calculate the combined margin (CM) emission factor.

Step 1: Identify the relevant electricity systems

As described in tool "For determining the electricity emission factors, identify the relevant project electricity system. Similarly, identify any connected electricity systems". It also states that "If the DNA of the host country has published a delineation of the project electricity system and connected electricity systems, these delineations should be used". Keeping this into consideration, the Central Electricity Authority (CEA), Government of India has divided the Indian Power Sector into five regional grids viz. Northern, Eastern, Western, North-eastern and Southern.

However, since August 2006, however, all regional grids except the Southern Grid had been integrated and were operating in synchronous mode, i.e., at same frequency. Consequently, the Northern, Eastern, Western and North-Eastern grids were treated as a single grid named as NEWNE grid from FY 2007-08 onwards for the purpose of this CO_2 Baseline Database. As of 31 December 2013, the Southern grid has also been synchronized with the NEWNE grid; hence forming one unified Indian Grid. Since the project supplies electricity to the Indian grid, emissions generated due to the electricity generated by the Indian grid as per CM calculations will serve as the baseline for this project.

[13] https://cea.nic.in/wp-content/uploads/2021/03/Baseline_Carbon_Dioxide_Emission_Database_Version_19.0.zip

Step 2: Choose whether to include off-grid power plants in the project electricity system (optional)

Project owner may choose between the following two options to calculate the operating margin and build margin emission factor:

Option I: Only grid power plants are included in the calculation.

Option II: Both grid power plants and off-grid power plants are included in the calculation.

The Project owner has chosen only grid power plants in the calculation.

Step 3: Select a method to determine the operating margin (OM)

The calculation of the operating margin emission factor ($EF_{grid,OM,y}$) is based on one of the following methods, which are described under Step 4:

(a) Simple OM; or

(b) Simple adjusted OM; or

(c) Dispatch data analysis OM; or

(d) Average OM.

The data required to calculate Simple adjusted OM and Dispatch data analysis OM is not possible due to lack of availability of data to project owner. The choice of other two options for calculating operating margin emission factor depends on generation of electricity from low-cost/must-run sources. In the context of the methodology low cost/must run resources typically include hydro, geothermal, wind, low-cost biomass, nuclear and solar generation.

Share of Must-Run (Hydro/Nuclear) (% of Net Generation)						
	2017-18	2018-19	2019-20	2020-21	2021-22	2022-23
India	14.3%	14.5%	17.0%	16.5%	15.8%	15.33%

Source: Central Electricity Authority (CEA) database Version 19, November '2023[14]

The above data clearly shows that the percentage of total grid generation by low-cost/ must-run plants (on the basis of average of five most recent years) for the Indian grid is less than 50 % of the total generation. Thus, the Average OM method cannot be applied, as low cost/must run resources constitute less than 50% of total grid generation.

The simple OM emission factor is calculated as the generation-weighted average CO_2 emissions per unit net electricity generation (tCO_2/MWh) of all generating power plants serving the system, not including low-cost/must-run power plants/units.

For the simple OM, the simple adjusted OM and the average OM, the emissions factor can be calculated using either of the two following data vintages:

(a) **Ex-ante option:** if the ex-ante option is chosen, the emission factor is determined during the registration validation stage, thus no monitoring and recalculation of the emissions factor during the crediting period is required.

OR

(b) **Ex-post option:** if the ex-post option is chosen, the emission factor is determined for the year in which the project activity displaces grid electricity, requiring the emissions factor to be updated annually during monitoring.

[14] https://cea.nic.in/wp-content/uploads/2021/03/Baseline_Carbon_Dioxide_Emission_Database_Version_19.0.zip

Project owner has chosen ex-ante option for calculation of Simple OM emission factor using a 3-year generation-weighted average, based on the most recent data available at the time of submission of the PD.

OM determined at registration stage will be the same throughout the crediting period. There will be no requirement to monitor & recalculate the emission factor during the crediting period.

Step 4: Calculate the operating margin emission factor ($EF_{grid,OMSimple,y}$) according to the selected method

The operating margin emission factor has been calculated using a 3-year data vintage:

Net Generation in Operating Margin (GWh) (incl. Imports)			
	2020-21	2021-22	2022-23
INDIAN Grid	958,218	1,035,672	1,117,846

Simple Operating Margin (tCO2/MWh) (incl. Imports)			
	2020-21	2021-22	2022-23
INDIAN Grid	0.9402	0.9605	0.971

Weighted Generation Operating Margin	
INDIAN Grid	**0.9580**

STEP 5: Calculate the build margin emission factor ($EF_{BM,y}$)

Option 1 as described above is chosen to calculate the build margin emission factor for the project activity. BM is calculated ex-ante based on the most recent information available at the time of submission of PD and is fixed for the entire crediting period.

Build Margin (tCO_2/MWh) (not adjusted for imports)	
	2022-23
INDIAN Grid	**0.8670**

STEP 6: Calculate the combined margin (CM) emissions factor

Combined Margin – The combined margin is the weighted average of the simple operating Margin and the build margin. In particular, for intermittent and non-dispatch able generation types such as wind and solar photovoltaic, Tool 07: Tool to calculate the emission factor for an electricity system, Version 7.0.0, EB 100, Annex 4, allows to weigh the operating margin and Build margin at 75% and 25%, respectively for wind and solar projects and 50% and 50%, respectively for hydro and biomass projects.

The baseline emission factor is calculated using the combined margin approach as described in the following steps:

Calculation of Baseline Emission Factor EF_y

The baseline emission factor **EF_y** is calculated as the weighted average of the Operating Margin emission factor (**$EF_{OM,y}$**) and the Build Margin emission factor (**$EF_{BM,y}$**):

$$EFy = w_{OM} * EF_{OM,y} + w_{BM} * EF_{BM,y}$$

Where,

w_{OM} 75% weight for solar energy projects

w_{BM} 25% weight for solar energy projects

$EF_{OM,y}$ calculated as described in Steps 3&4 above (tCO_2/MWh)

$EF_{BM,y}$ calculated as described in Steps 5 above (tCO_2/MWh)

Baseline Emission factor (INDIAN Grid) = 0.75*0.9580 + 0.25*0.8670

= 0.9352 tCO_2/MWh

The baseline emission factor is ex-ante parameter and will remain constant throughout the crediting period.

$EF_{grid,CM,y}$	**= Combined Margin Grid Emission Factor = 0.9352 tCO_2/MWh**

4.2 Project Emissions

As per the approved consolidated Methodology ACM0002, "Grid-connected electricity generation from renewable sources" (Version 21.0) para 35: "For most renewable energy power generation project activities, PE_y = 0. However, some project activities may involve project emissions that can be significant. These emissions shall be accounted for as project emissions by using the following equation:

$$PE_y = PE_{FF,y} + PE_{GP,y} + PE_{HP,y} + PE_{BESS,y}$$

Where,

PE_y = Project emissions in year y ($t\ CO_2e/yr$)

$PE_{FF,y}$ = Project emissions from fossil fuel consumption in year y (tCO_2/yr)

$PE_{GP,y}$ = Project emissions from the operation of dry, flash steam or binary geothermal power plants in year y (tCO_2e/yr)

$PE_{HP,y}$ = Project emissions from water reservoirs of hydro power plants in year y (tCO_2e/yr)

$PE_{BESS,y}$ = Project emissions from charging of a BESS using electricity from the grid or from fossil fuel electricity generators (tCO_2e/yr)

As the project activity is the installation of a new grid-connected solar Power plant and does not involve any project emissions from fossil fuel, operation of dry, flash steam or binary geothermal power plants, and from water reservoirs of hydro power plants. Therefore $PE_{FF,y}$, $PE_{GP,y}$, $PE_{HP,y}$, $PE_{BESS,y}$ is equal to zero and thus, PE_y = 0.

4.3 Leakage Emissions

No other leakage emissions are considered. The emissions potentially arising due to activities such as power plant construction and upstream emissions from fossil fuel use (e.g.: extraction, processing, transport etc.) are neglected.

4.4 Estimated GHG Emission Reductions and Carbon Dioxide Removals

As per the paragraph 62 of the methodology ACM0002 "Grid-connected electricity generation from renewable sources" Version 21.0 emission reductions are calculated as follows

Emission Reductions

$ER_y = BE_y - PE_y$

Where,

ER_y = Emission reductions in year *y* (tCO_2e/yr)
BE_y =Baseline emissions in year *y* (tCO_2/yr)
PE_y =Project emissions in year y (tCO_2/yr)

Baseline Emissions (BE_y):

As per the above sections 4.1 Baseline Emission (BE_y) will be calculated as per the below equation

$$BE_y = EG_{facility,\, y} \times EF_{grid,CM,y}$$

Where,

ER_y = Emission Reduction in tCO_2/year
BE_y = Baseline emission in tCO_2/year
$EG_{facility,y}$ = Total quantity of net electricity delivered to the INDIAN grid in year y (MWh/yr)
$EF_{grid,CM,y}$ = Baseline grid emission factor (t CO_2/MWh)
= 0.9310 tCO_2/MWh

Hence the baseline emission for the entire crediting period of the Project is calculated as below:

Crediting Period	Total Net Electricity generation	Baseline emissions (tCO_2 e)
Year		
12-Apr-2024 to 31-Dec-2024	130,641	122,175
01-Jan-2025 to 31-Dec-2025	179,719	168,073
01-Jan-2026 to 31-Dec-2026	178,820	167,233
01-Jan-2027 to 31-Dec-2027	177,926	166,397
01-Jan-2028 to 31-Dec-2028	177,037	165,565
01-Jan-2029 to 31-Dec-2029	176,151	164,737
01-Jan-2030 to 31-Dec-2030	175,271	163,913
01-Jan-2031 to 11-Apr-2031	48,257	45,130
Total	**1,243,822**	**1,163,222**

The baseline emission reduction for the entire crediting period of the Project is

$BE_y = EG_{facility,\, y} * EF_{grid,CM,y}$ = 1,243,822 MWh x 0.9352 tCO_2/MWh = **1,163,222 tCO_2**

$$BE_y = 1,163,222\ tCO_2$$

Project Emissions (PE_y):

As explained in the section 4.2 Project emissions from the project is considered Zero.

$$PE_y = 0$$

Net Emission Reductions (ER_y):

$ER_y = BE_y - PE_y$

Since the project emissions are estimated as zero

$ER_y = BE_y$

Considering the commissioning date of the project and annual degradation, the emission reduction estimation for the entire crediting period is provided in the below section.

Vintage period	Estimated baseline emissions (tCO_2e)	Estimated project emissions (tCO_2e)	Estimated leakage emissions (tCO_2e)	Estimated reduction VCUs (tCO_2e)	Estimated removal VCUs (tCO_2e)	Estimated total VCUs (tCO_2e)
12-Apr-2024 to 31-Dec-2024	122,175	0	0	122,175	0	122,175
01-Jan-2025 to 31-Dec-2025	168,073	0	0	168,073	0	168,073
01-Jan-2026 to 31-Dec-2026	167,233	0	0	167,233	0	167,233
01-Jan-2027 to 31-Dec-2027	166,397	0	0	166,397	0	166,397
01-Jan-2028 to 31-Dec-2028	165,565	0	0	165,565	0	165,565
01-Jan-2029 to 31-Dec-2029	164,737	0	0	164,737	0	164,737
01-Jan-2030 to 31-Dec-2030	163,913	0	0	163,913	0	163,913
01-Jan-2031 to 11-Apr-2031	45,130	0	0	45,130	0	45,130
Total	**1,163,222**	**0**	**0**	**1,163,222**	0	**1,163,222**

5 MONITORING

5.1 Data and Parameters Available at Validation

Data / Parameter	$EF_{grid,OM,y}$
Data unit	tCO_2/MWh
Description	Operating Margin CO_2 emission factor in year y of Indian Grid.
Source of data	Calculated from CEA database, Version 19, November 2023[15]
Value applied	0.9580
Justification of choice of data or description of measurement methods and procedures applied	***Not Applicable*** Type of meter: Not Applicable Location of meter: Not Applicable Accuracy of meter: Not Applicable Serial number of meter: Not Applicable Calculated in line with "*Tool to calculate the emission factor for an electricity system*" version 7.0 using data from Central Electricity Authority of India's (CEA) "*Baseline Carbon Dioxide Emission Database Version 19.0*". The value used is calculated ex-ante as generation based weighted average of last three years of the operating margin provided in the CEA database. Weighted average = $\sum_{i=1 \text{ to } n}$ (Net generation in operating margin in year i * Simple operating margin in year i)/$\sum_{i=\ 1 \text{ to } n}$ (Net generation in operating margin of year i)
Purpose of data	Baseline Emission calculation
Comments	The operating margin emission factor is a 3-year generation-weighted average (2020-23). The operating Margin is calculated ex ante and fixed during the crediting period

Data / Parameter	$EF_{grid,BM,y}$
Data unit	tCO_2/MWh
Description	Build Margin CO_2 emission factor in year y of Indian Grid.
Source of data	Calculated from CEA database, Version 19, November 2023
Value applied	0.8670
Justification of choice of data or description of	***Not Applicable*** Type of meter: Not Applicable

[15] https://cea.nic.in/wp-content/uploads/2021/03/Baseline_Carbon_Dioxide_Emission_Database_Version_19.0.zip

measurement methods and procedures applied	<table><tr><td>Location of meter</td><td>Not Applicable</td></tr><tr><td>Accuracy of meter</td><td>Not Applicable</td></tr><tr><td>Serial number of meter</td><td>Not Applicable</td></tr></table> Calculated in line with "*Tool to calculate the emission factor for an electricity system*" version 7.0 using data from Central Electricity Authority of India's (CEA) "*Baseline Carbon Dioxide Emission Database Version 19.0*". The value used is calculated ex-ante as generation based weighted average of last three years of the operating margin provided in the CEA database.
Purpose of Data	Baseline Emission calculation
Comments	The Build Margin would be calculated ex ante and fixed during the crediting period. For ex ante calculation the most recent data (2022-23) available has been used and the build margin is thus calculated.

Data / Parameter	$EF_{grid,CM,y}$
Data unit	tCO_2/MWh
Description	Combined Margin CO_2 emission factor in year y of Indian Grid.
Source of data	Calculated from CEA database, Version 19, November 2023
Value applied	0.9352
Justification of choice of data or description of measurement methods and procedures applied	<table><tr><td colspan="2">***Not Applicable***</td></tr><tr><td>Type of meter</td><td>Not Applicable</td></tr><tr><td>Location of meter</td><td>Not Applicable</td></tr><tr><td>Accuracy of meter</td><td>Not Applicable</td></tr><tr><td>Serial number of meter</td><td>Not Applicable</td></tr></table> The data has been considered in accordance to the Tool 07: Tool to calculate emission factor of an electricity system. The tool guides to take 75% weightage of EF_{grid}, OM_{simple}, & 25% weightage of $EF_{grid,BM,y}$.
Purpose of Data	Baseline Emission calculation
Comments	The combined margin would be calculated ex-ante and fixed for the entire crediting period and the combined margin thus calculated is 0.9352.

5.2 Data and Parameters Monitored

Data / Parameter	$EG_{facility,y}$
Data unit	MWh/Year

<table>
<tr><td>Description</td><td>Quantity of net electricity generation supplied by the project (Solar) plant/unit to the grid in year y</td></tr>
<tr><td>Source of data</td><td>JMR from End user (consumer)</td></tr>
<tr><td>Description of measurement methods and procedures to be applied</td><td>Measured & calculated</td></tr>
<tr><td>Frequency of monitoring/recording</td><td>Monthly recording</td></tr>
<tr><td>Value applied</td><td>1,243,822</td></tr>
<tr><td>Monitoring equipment</td><td>
<table>
<tr><th>Project 1</th><th>Main Meter</th><th>Check meter</th></tr>
<tr><td>Type of meter</td><td colspan="2">Trivector Bi-directional meters (Manufacturer: Secure Meters Limited)</td></tr>
<tr><td>Location of meter</td><td colspan="2">Substation</td></tr>
<tr><td>Accuracy of meter</td><td colspan="2">0.2s</td></tr>
<tr><td>Serial number of meter</td><td colspan="2">To Be Provided</td></tr>
<tr><td>Calibration frequency</td><td colspan="2">To Be Provided</td></tr>
<tr><td>Date of Calibration/ validity</td><td colspan="2">To Be Provided</td></tr>
<tr><td>Reference No. of Calibration Certificate</td><td colspan="2">To Be Provided</td></tr>
<tr><td>Calibration Status</td><td colspan="2">To Be Provided</td></tr>
</table>
Energy meters of accuracy class 0.2s.
Main and Check meters are installed at the Saktapur Substation by the electricity utility to measure the net exported electricity from the plant.
</td></tr>
<tr><td>QA/QC procedures to be applied</td><td> The meter(s) shall be calibrated and maintained by the DISCOMs as per their own schedule, and this frequency of meter calibration is not within the control of the Project Owner.
Calibration of electricity meters is carried out in-line with the National standard[16] which recommends at least once in 5-year calibration or whenever abnormal difference/inconsistency is observed between main meter and check meter. </td></tr>
</table>

[16] (Page number 18 of https://cea.nic.in/old/reports/regulation/CEA_metering_regulation_amendment_2019.pdf)

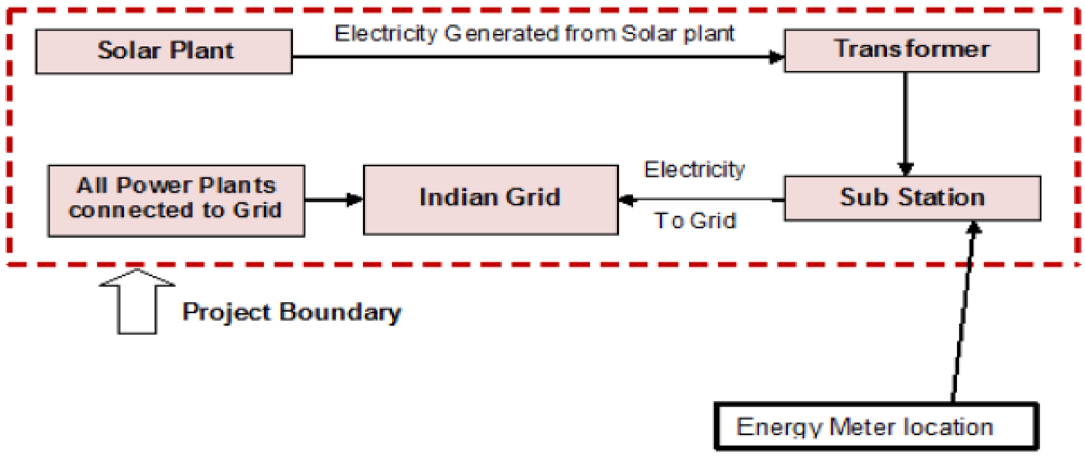
Purpose of data	Baseline Emission Calculations
Calculation method	The value of net electricity generation supplied to the grid as per Monthly Joint Meter Reading Report forms the basis for calculation of the emission reductions, which can be cross-checked from the invoice raised to Consumer. The Net electricity is calculated based on Export - import. Monthly meter readings are taken from the main and check meter installed at metering point and certified by the representatives of DISCOM Officials and the representatives of the project owner participant. The export and import values of the monthly Joint Meter Reports is cross checked with the export and import values mentioned in the invoice.
Comments	Monitored data will be stored for at least two years after the end of crediting period or the last issuance of this project activity, whichever occurs later

5.3 Monitoring Plan

The monitoring plan is developed in accordance with the modalities and procedures for VERRA project activities and is proposed for grid-connected solar power project being implemented in Madhya Pradesh, India. The monitoring plan, which will be implemented by the project owner describes about the monitoring organization, parameters to be monitored, monitoring practices, quality assurance, quality control procedures, data storage and archiving.

The project owner has entered into agreement with the O&M Contractor for the operation and maintenance of Solar Plant. O&M contractor will provide a monthly report, which includes generation data, major breakdown events and machine availability. Project manager is responsible for recording of monthly meter readings of export and import. Monthly power export and import data will be sent regularly to site in charge of each project separately.

The electricity generated from the project activity will be supplied to the national grid for the complete project lifetime under a long-term PPA with MPPMCL. The electricity generated from the project activity before entering the grid at the connection point will be measured by digital kilowatt hour (kWh) meters. The power meters at the connection point are bi-directional nature which records the net export and import. Both main and check meters of +/-0.2s accuracy class have been installed at the substation. These are the 3-phase bi-directional trivector meters.



The steps of monitoring the electricity supplied to the national grid and the electricity imported from grid by the project are as follows:

1. The electricity supplied by the project to the grid and electricity imported from the grid are measured automatically by the bi-directional meter systems (main and backup power meters). The data is measured continuously

2. Persons in charge of data record and meter supervisor from the Project owner together with staff from DISCOM shall read and collect data from power meters on the first day of every month, the result will be signed by both parties and kept respectively.

3. The data from the backup power meters will be cross checked with the data from main power meter. The data from back-up system will be used in case of failing of the main meter;

4. The Project owner provides the record of main, backup power meters, copy of invoices and other related documents to the verifier.

Personal Training: The project employed qualified and experienced persons for plant operation. The training period shall be for three months, as this would be adequate and necessary to ensure proper imparting of the objective. The training course will be thoroughly and meticulously designed, highlighting the objectives, salient features, operational aspects and trouble shooting.

Data recording & archiving: The project owner shall maintain data both in electronic form and hard copies.

Emergency preparedness:

In case of any unforeseen event that is not covered under this monitoring plan, staff of the operation division will immediately inform the chief of the operation division. The chief of the operation division is then responsible to ensure that the cause for the unforeseen event is detected, the event is remedied and for the period of time in which the unforeseen event has occurred uncertainty in data gathered is limited as much as possible.

- In normal condition, the data of main power meter will be used as the basis of payment by the DISCOM to the project owner and to calculate the emission reductions by the project activity.
- In case the main power meter is in failure and the backup power meter is still in good operation, the result of the backup meter will be used to calculate the emission reduction by the project activity.

- In case of both main and backup power meters are in failure, the Project Owner and the power company (DISCOM) will jointly calculate a conservative estimate of power supplied to the grid. The assumptions used to estimate net electricity supply to the grid will be signed by both a representative of the project owner as well as a representative of the power company (DISCOM).
- In case any power meters are in failure, the Project Owner will inform DISCOM immediately and contract with the authorized party (belong to DISCOM) to verify/ calibrate and/or replace the failed power meter.

Internal auditing

The main energy meter readings are verified against the reading of the backup meter to check any considerable variation. In case, considerable variation found, then either main meter reading or check meter reading whichever lower will be considered. The electricity readings recorded are used for electricity invoicing to consumer. Hence, the data collected is of high accuracy and authorized by DISCOM.

APPENDIX 1: COMMERCIALLY SENSITIVE INFORMATION

Not Applicable

Section	Information	Justification
Not Applicable	Not Applicable	Not Applicable